Caio Barbosa Rodrigues

Doctoral student
Institute for Pure and Applied Mathematics (IMPA), Rio de Janeiro, Brazil
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Research Interests

Differential geometry and geometric analysis, with a focus on Willmore surfaces, minimal surfaces, and bifurcation theory.

Education

Mar. 2023–present	**Doctoral studies in Mathematics** IMPA. Advisor: Prof. Lucas Ambrozio. Expected thesis defense: March 2027. Thesis: *Bifurcations of the Clifford torus as Willmore surfaces in Berger spheres*.
2020 (2nd sem.)– 2022 (2nd sem.)	**Master's degree in Pure Mathematics** IMPA. Advisor: Prof. Vinicius Gripp Barros Ramos.
2018 (1st sem.)– 2020 (1st sem.)	**Bachelor's degree in Mathematics** University of Brasília (UnB). Undergraduate research: *Groups with a finite number of centralizers*. Advisor: Prof. Igor dos Santos Lima.
2015 (2nd sem.)– 2018 (1st sem.)	Earlier studies: Aerospace Engineering, UnB; transferred to Mathematics.

Publications and Preprints

1. Caio B. Rodrigues. *Bifurcations of the Clifford Torus as Willmore Surfaces in Berger Spheres*. Preprint, 2026. arXiv:2607.02468. Submitted to a journal.
2. Igor Lima and Caio Rodrigues. *On solubility of groups with finitely many centralizers*. International Electronic Journal of Algebra **32** (2022), 241–245. doi:10.24330/ieja.1144159.

Invited Talks

6 Oct. 2026	**Bifurcations of the Clifford Torus as Willmore Surfaces in Berger Spheres** Differential Geometry Seminar, IMPA.

Poster Presentations

Willmore Bifurcations of the Clifford Torus in Berger Spheres

13–17 July 2026	22nd School of Differential Geometry, Teresina, Brazil.
22 June– 3 July 2026	European School of Differential Geometry, L'Aquila, Italy.
9–13 Mar. 2026	XI Workshop on Differential Geometry, Maceió, Brazil.

Teaching Experience

Jan.–Feb. 2025 and 2026	**Teaching Assistant: Differential Geometry of Curves and Surfaces** IMPA Summer School; master's-level course. Led problem-solving and review sessions.
2016 (1st sem.)	**Teaching Assistant: Calculus I** UnB; undergraduate course. Led problem-solving and review sessions.
2016	**Tutor: Physics I** UnB; undergraduate course. Provided one-to-one tutoring for a student with central auditory processing disorder.

Scholarships

Mar. 2023– Mar. 2027	**Doctoral scholarship** Funding agency: National Council for Scientific and Technological Development (CNPq), Brazil.
2020 (2nd sem.)– 2022 (2nd sem.)	**Master's scholarship** Funding agency: Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES), Brazil.
2018	**Undergraduate research scholarship** Funding agency: CNPq.

Languages

Portuguese — native.
English — proficient.